

Last Name:

First Name:

ID:

Question 1.

5 pts

In a colorful toy box, there are 3 unique toy cars: 1 red car (R), 1 green car (G), and 1 blue car (B). You decide to play a game where you randomly pick 1 toy car, examine its design, and then either return it to the box or not before picking a second car.

- (2.5 pts) You pick 1 toy car, return it to the box, and then pick a second toy car. Describe the sample space of your selections.
- (2.5 pts) This time, you pick the first toy car and **do not** return it to the box before picking the second one. Describe the new sample space.

Solution 1.

- The sample space consists of the possible experimental outcomes, which in this case is given by

$$\{(R, R), (R, G), (R, B), (G, R), (G, G), (G, B), (B, R), (B, G), (B, B)\}.$$

- If the first car is not replaced then our sample space loses all “paired” terms in the above (i.e. terms like (R, R)) and it becomes

$$\{(R, G), (R, B), (G, R), (G, B), (B, R), (B, G)\}.$$

Question 2.

7 pts

- (3.5 pts) A book club has 15 members. If they want to select 5 members to read a book together, how many different groups can be formed?
- (3.5 pts) How many distinct ways can the letters in the word “BANANA” be arranged?

Solution 2.

- Using the combination formula:

$$\binom{15}{5} = \frac{15!}{5!(15-5)!} = \frac{15!}{5! \cdot 10!} = \frac{15 \times 14 \times 13 \times 12 \times 11}{5 \times 4 \times 3 \times 2 \times 1} = \frac{360360}{120} = 3003$$

- The word “BANANA” has 6 letters where A appears 3 times and N appears 2 times:

$$\text{Number of arrangements} = \frac{n!}{n_1! \cdot n_2! \cdot \dots} = \frac{6!}{3! \cdot 2! \cdot 1!} = \frac{720}{6 \times 2} = 60$$

Question 3.

8 pts

- i. (4 pts) Suppose that a person chooses a letter at random from R E S E R V E and then chooses one at random from V E R T I C A L. What is the probability that the same letter is chosen?
- ii. (4 pts) A jar contains 12 cookies, 3 of which are burnt. What is the probability that if we randomly select 4 cookies from the jar, exactly 2 of them will be burnt?

Solution 3.

- i. (4 pts) The same letter could be chosen if and only if it comes from one of R, E or V. The probability of R is chosen from both words is

$$\frac{2}{7} \cdot \frac{1}{8} = \frac{2}{56}.$$

The probability of E is chosen from both words is

$$\frac{3}{7} \cdot \frac{1}{8} = \frac{3}{56}.$$

Finally the probability of V is chosen from both words is

$$\frac{1}{7} \cdot \frac{1}{8} = \frac{1}{56}.$$

So the total probability is the sum of all the above probabilities $\frac{3}{28}$.

- ii. (4 pts) Define the event $A := \{\text{we pick 2 burnt cookies and 2 not}\}$, then

$$\begin{aligned} P(A) &= \frac{\binom{3}{2} \binom{9}{2}}{\binom{12}{4}} \\ &= \frac{(3)(36)}{495} \\ &= 0.2182. \end{aligned}$$

Question 4.

7 pts

Given that $P(A) = 0.4$, $P(B^c) = 0.5$, and $P(A \cap B) = 0.2$, find the following:

- i. (1 pts) $P(A^c) =$
- ii. (1 pts) $P(A \cup B) =$
- iii. (1 pts) $P(A|B) =$
- iv. (1 pts) $P(A \cap B^c) =$
- v. (2 pts) Are A and B independent? Why?
- vi. (1 pts) Are A and B mutually exclusive? Why?

Solution 4.

- i. $P(A^c) = 1 - P(A) = 1 - 0.4 = 0.6$
- ii. $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + (1 - 0.5) - 0.2 = 0.7.$
- iii. $P(A|B) = P(A \cap B)/P(B) = 0.2/0.5 = 0.4.$
- iv. $P(A \cap B^c) = P(A) - P(A \cap B) = 0.4 - 0.2 = 0.2.$
- v. Since $P(A \cap B) = 0.2 = P(A)P(B) = (0.4)(0.5)$, then, A and B are independent.
- vi. No, since $P(A \cap B) \neq 0.$

Question 5.

8 pts

Sixty percent of the students at a certain school wear neither a ring nor a necklace. Twenty percent wear a ring and 30 percent wear a necklace. If one of the students is chosen randomly, what is the probability that this student is wearing

- i. (4 pts) a ring or a necklace?
- ii. (4 pts) a ring and a necklace?

Solution 5.

Let $P(A)$ be the probability that a student wears a ring. Let $P(B)$ be the probability that a student wears a necklace. Then from the information given we have that

$$P(A) = 0.2, \quad P(B) = 0.3, \quad P((A \cup B)^c) = 0.6$$

- i. We desire to calculate for this $P(A \cup B)$, which is given by

$$P(A \cup B) = 1 - P((A \cup B)^c) = 1 - 0.6 = 0.4$$

- ii. We desire to calculate $P(A \cap B)$, which can be calculated by

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.2 + 0.3 - 0.4 = 0.1$$

Question 6.

(+5 pts)

- i. (2.5 pts) Show that the probability that exactly one the events E or F occurs equals to $P(E) + P(F) - 2P(E \cap F)$.
- ii. (2.5 pts) Prove that $P(E \cap F^c) = P(E) - P(E \cap F)$.

Solution 6.

- i. (2.5 pts)

$$P(\text{Exactly one of } E \text{ or } F) = P(E \Delta F) = P(E \cup F) - P(E \cap F)$$

Now, recall the formula for the union of two events:

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

Substituting into the expression:

$$\begin{aligned}P(E \triangle F) &= [P(E) + P(F) - P(E \cap F)] - P(E \cap F) \\&= P(E) + P(F) - 2P(E \cap F)\end{aligned}$$

Therefore,

$$\boxed{P(\text{Exactly one of } E \text{ or } F) = P(E) + P(F) - 2P(E \cap F)}$$

ii. We begin by observing that the event E can be partitioned into two disjoint subsets:

$$E = (E \cap F) \cup (E \cap F^c)$$

Since $(E \cap F)$ and $(E \cap F^c)$ are disjoint events, we can apply the additive rule of probability:

$$P(E) = P(E \cap F) + P(E \cap F^c)$$

Rearranging the equation:

$$P(E \cap F^c) = P(E) - P(E \cap F)$$

Therefore,

$$\boxed{P(E \cap F^c) = P(E) - P(E \cap F)}$$